

```

-----
ModuleNotFoundError                                Traceback (most recent call last)
Cell In[1], line 1
----> 1 from notebook.services.config import ConfigManager
      2 cm = ConfigManager()
      3 cm.update('liverveal', {
      4     'width': 1920,
      5     'height': 1080,
      6     'scroll': True,
      7 })

ModuleNotFoundError: No module named 'notebook.services'

```

Week 06: Monday, AST 5011: Astrophysical Systems

Stellar Modelling and Astroseismology

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

The Equations of Stellar Structure

We begin with the need to construct a model that describes the pressure, density, temperature and luminosity and their derivatives:

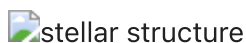
$$P = P(\rho, T, \mathbf{X})$$

$$\mathbf{E} = \mathbf{E}(\rho, \mathbf{T}, \mathbf{X})$$

$$\kappa = \kappa(\rho, \mathbf{T}, \mathbf{X})$$

$$\epsilon = \epsilon(\rho, \mathbf{T}, \mathbf{X})$$

where $\mathbf{X}$ is shorthand for composition (as in a specification of nuclear species).



stellar structure

We also have our various ∇ relations leading to the creation of a fourth-order differential equation in space or mass requiring 4 boundary conditions.

How to solve these equations is not trivial. But we can make some simplifications that lead to the use of Polytropes. **Polytropes** are pseudo-stellar models for which power

law equations of pressure versus density are assumed a priori but where no reference to heat transfer or thermal balance is made.

Polytropic Equations of State and Polytropes

A polytropic stellar model is defined as

$$P(r) = K\rho^{1+1/n}(r)$$

where n is the *polytropic index* and K is a constant of proportionality.

From Eqn 7.16–7.26 lead to the *Lane-Emden* equation,

a dimensionless form of Poisson's equation where ξ is a dimensionless radius and θ relates to the density.

Models corresponding to solutions of this equation for a chosen n are called “polytropes of index n ” and the solutions themselves are “Lane–Emden solutions” and are denoted by $\theta_n(\xi)$.

Solutions to the LE equation are often restricted to $0 \lesssim n \lesssim 5$.

Some relevant examples for Polytropes

1. The pressure of the completely degenerate but nonrelativistic electron gas goes as $\rho^{5/3}$. Hence, by the definition of the polytropic equation of state (7.16), n for this case is 1.5 (or “a three-halves polytrope”).
2. The density exponent for the fully relativistic case is $4/3$ and thus $n = 3$ (or “an n equal three polytrope”).
3. Recall that $P \propto \rho^{5/3}$ in an ideal gas convection zone. If no ionization is taking place (almost a contradiction for a real convection zone) then $\Gamma_2 = 5/3$ and $n = 3/2$ again.

These represent common scenarios for polytropes and rely on the conditions to which we are comparing.

Solving the Stellar Structure Equations

Thorough discussion in [MESA 1 Section 6](#) for more details.

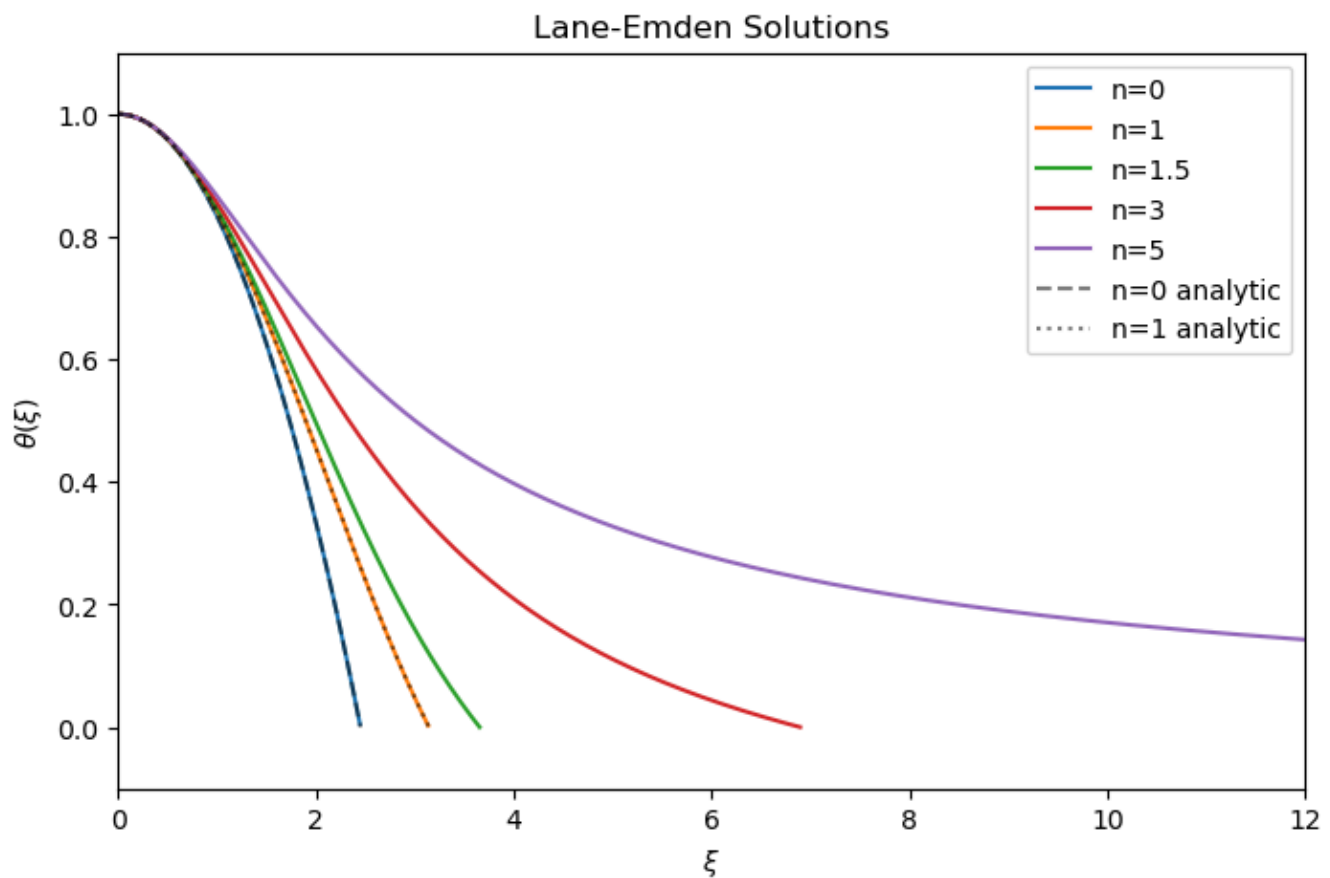
In-Class Exercise: Lane-Emden Solutions

Goal

Build intuition for how the polytropic index n affects stellar density structure by solving the Lane-Emden equation numerically.

Task

1. Complete the right-hand side of the Lane-Emden equation.
2. Solve for $n = 0, 1, 1.5, 3, 5$ and plot $\theta(\xi)$ for each.
3. Overplot the analytic solutions for $n = 0$ and $n = 1$:
 - $n = 0$: $\theta = 1 - \xi^2/6$, with $\xi_1 = \sqrt{6}$
 - $n = 1$: $\theta = \sin(\xi)/\xi$, with $\xi_1 = \pi$
4. Which polytrope has the most centrally concentrated density profile?



► Solution

Radial Stellar Pulsations

Stellar pulsations start from *pressure waves* (sound waves that resonate in the stellar interior).

These radial oscillations act like standing waves with a node at the center and open end at the surface.

Fundamental mode - one node at the center

First overtone - one node at the center and one between center and surface

Second overtone - one node at the center and two between center and surface

Most radially pulsating stars, *Cepheids*, are oscillating in their fundamental mode.

Adiabatic Oscillations

In practice, oscillations will be about some equilibrium structure at equilibrium radius r_0 and mass m . Under these assumptions, the radial oscillation frequency is found to be:

$$\omega^2 = (3\gamma_{\text{ad}} - 4) \frac{Gm}{r_0^3}$$

Recall $\gamma_{\text{ad}} \equiv \Gamma_1$.

For $\gamma_{\text{ad}} > 4/3$, we have $\omega^2 > 0$ and **dynamical stability**.

For $\gamma_{\text{ad}} < 4/3$, we have $\omega^2 < 0$, indicating exponential growth of the perturbations and **dynamical instability**.

If we average ω over the entire star, we can get an estimate for the **pulsation frequency of the fundamental mode**:

$$\Pi_0 = \frac{2\pi}{\sqrt{(3\gamma_{\text{ad}} - 4) \frac{GM}{R^3}}} \equiv \left(\frac{3\pi}{(3\gamma_{\text{ad}} - 4) G \bar{\rho}} \right)^{1/2}$$

where we have used the total radius and mass of the star.

We can generalize this to:

$$\Pi = Q \left(\frac{\bar{\rho}}{\rho_{\odot}} \right)$$

We have $Q \approx 0.04$ days for the fundamental mode and smaller for higher overtones.

In-Class Exercise: Pulsation Period from Mean Density

Goal

Use the period–mean density relation to estimate fundamental pulsation periods for different types of variable stars.

Task

1. Compute the mean density $\bar{\rho} = 3M/(4\pi R^3)$ for each star.

2. Compute the fundamental period $\Pi_0 = \left(\frac{3\pi}{(3\gamma_{\text{ad}}-4)G\bar{\rho}} \right)^{1/2}$ assuming $\gamma_{\text{ad}} = 5/3$.
3. Plot period versus mean density.
4. Do the computed periods match the observed ranges?

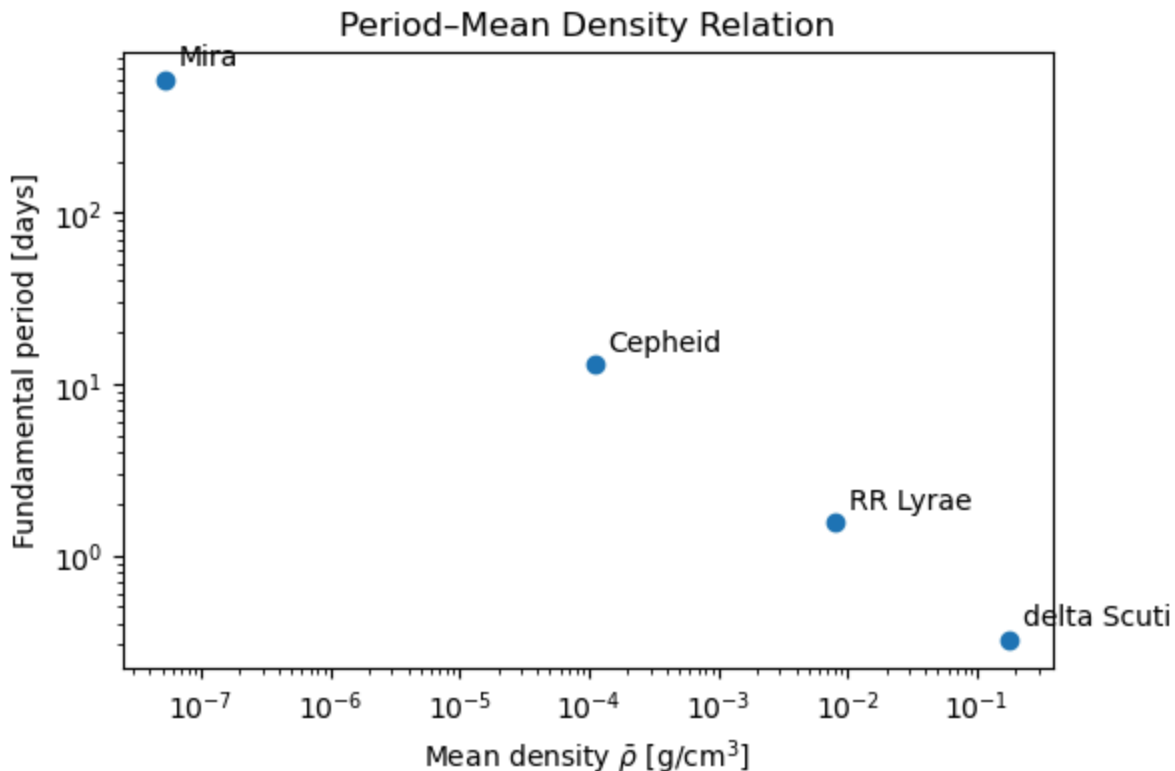
Star type	Observed period range
δ Scuti	0.02 – 0.25 days
RR Lyrae	0.2 – 1.0 days
Cepheid	1 – 100 days
Mira	100 – 700 days

delta Scuti: $\rho_{\text{bar}} = 1.803\text{e-}01 \text{ g/cm}^3$, $\Pi_0 = 0.32 \text{ days}$

RR Lyrae: $\rho_{\text{bar}} = 7.887\text{e-}03 \text{ g/cm}^3$, $\Pi_0 = 1.55 \text{ days}$

Cepheid: $\rho_{\text{bar}} = 1.100\text{e-}04 \text{ g/cm}^3$, $\Pi_0 = 13.12 \text{ days}$

Mira: $\rho_{\text{bar}} = 5.216\text{e-}08 \text{ g/cm}^3$, $\Pi_0 = 602.40 \text{ days}$



► Solution

Driving and damping of pulsations

ϵ -mechanism:

If the compressed region occurs where nuclear reactions occur, T rises and thus the nuclear energy generation rate. This could satisfy the criterion to grow the pulsation. Always taking place in the cores, but very small amplitudes. Might be important for very massive stars, not so for Cepheids.

κ -mechanism:

If the compressed layer becomes more opaque, heat is trapped. This causes an increase in P and T , which drive expansion and release of the heat as a result of the reduction of opacity. This can maintain the pulsation cycle and drive large scale pulsations.

We can write a requirement for this mechanism as:

$$\left(\frac{d \ln \kappa}{d \ln P} \right)_{\text{ad}} = \left(\frac{\partial \ln \kappa}{\partial \ln P} \right)_T + \left(\frac{\partial \ln \kappa}{\partial \ln T} \right)_P \nabla_{\text{ad}} \equiv \kappa_P + \kappa_T \nabla_{\text{ad}}$$

for pulsations via this mechanism we require $\kappa_P + \kappa_T \nabla_{\text{ad}} > 0$.

Pathways to Instability

1. $\kappa_T > 0$, occurs when H^- opacity dominates in very cool stars with $T < 10^4$ (K). Cepheids are too hot for this.
2. Small ∇_{ad} for a Kramers like opacity. We found that ∇_{ad} can be reduced in *partial ionization zones*.

Two key partial ionization zones

- $T \approx 1.5 \times 10^4$ - where $H \rightarrow H^+ + e^-$ and $He \rightarrow He^+ + e^-$ can occur.
- $T \approx 4 \times 10^4$ - where helium becomes doubly ionized $He^+ \rightarrow He^{++} + e^-$ (more opaque)

These considerations determine the location of the instability strip.

In-Class Exercise: Kappa-Mechanism Driving Condition

Goal

Explore when the opacity-driven pulsation condition is satisfied and understand why partial ionization zones are essential.

Background

For Kramers opacity $\kappa \sim \rho T^{-3.5}$, using an ideal gas ($P \propto \rho T$):

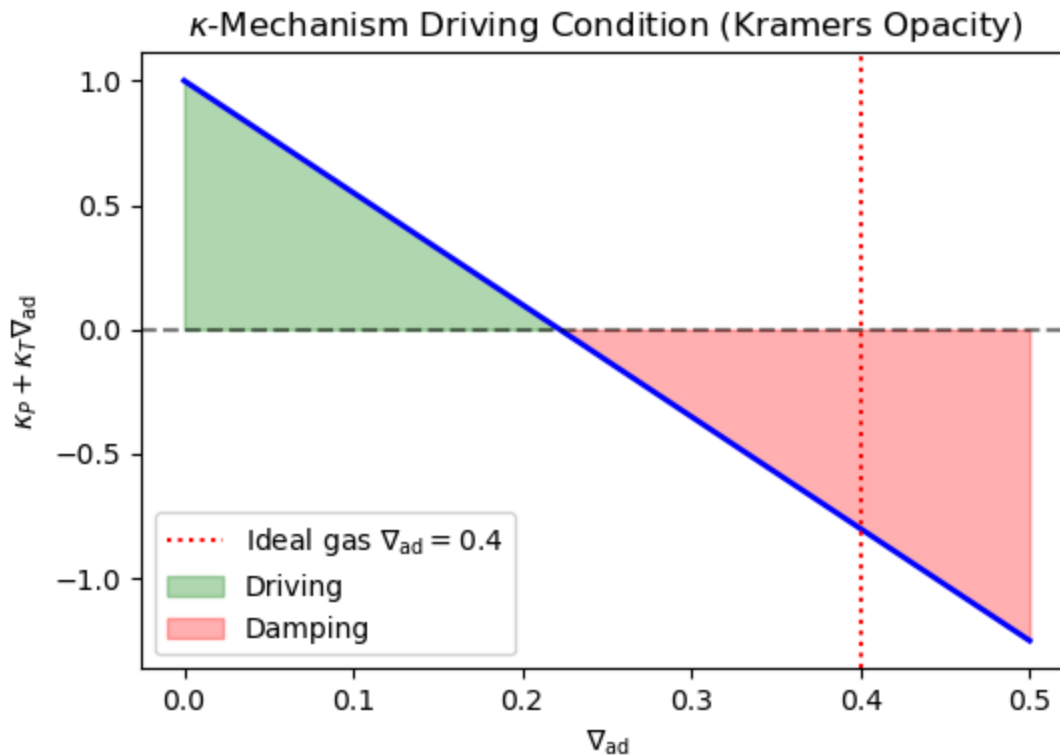
- At constant T : $\rho \propto P$, so $\kappa \propto P \cdot T^{-3.5}$, giving $\kappa_P = 1$
- At constant P : $\rho \propto 1/T$, so $\kappa \propto T^{-4.5}$, giving $\kappa_T = -4.5$

The driving condition is $\kappa_P + \kappa_T \nabla_{\text{ad}} > 0$.

Task

1. Evaluate the driving condition as a function of ∇_{ad} .

2. Find the critical ∇_{ad} below which driving occurs.
3. Mark where the ideal gas value ($\nabla_{\text{ad}} = 0.4$) falls.
4. Is driving possible in normal stellar material? What about partial ionization zones?



Driving requires $\nabla_{\text{ad}} < 0.222$

Ideal gas $\nabla_{\text{ad}} = 0.4$: condition = -0.80 (damped)

Partial ionization $\nabla_{\text{ad}} \sim 0.1$: condition = 0.55 (driving!)

► Solution

Example: Cepheids

The main gas involved is thought to be helium. The cycle is driven by the fact doubly ionized helium, the form adopted at high temperatures, is more opaque than singly ionized helium.

1. Outer layer is compressed
2. This layer is heated, helium is doubly ionized
3. The increased heat absorbs enough heat to expand
4. Cools and Helium is back to singly ionized (less opaque)

Cepheid variables become dimmest during the part of the cycle when the helium is doubly ionized.



polaris

A series of images of the pole star, Polaris, which is a Cepheid type variable. 4 frames taken at 24 hour intervals covering Polaris' approximately 4 day cycle during which its brightness varies by 0.27 magnitudes. **Credit: (Tim Wetherell 2022)**

Modern Pulsation Codes

GYRE by R. H. D. Townsend, S. A. Teitler - an open-source stellar oscillation code based on a new Magnus Multiple Shooting scheme. Paper [here](#). Solves the stellar pulsation equations (both adiabatic and non-adiabatic) using a novel Magnus Multiple Shooting numerical scheme. Integrated into MESA since MESA Paper II.

RSP MESA [module](#) - model the time evolution of large amplitude, self-excited, nonlinear pulsations over many cycles to produce luminosity and radial velocity histories that can be compared to observations.

In-Class Assignment: Comparing Polytropes to MESA models

Learning Objectives

- explore the relationship between lane-emen solutions and physical variables
- identify physics informed polytrope profiles for models of massive stars and white dwarfs

Lane-Emden polytrope equation

Derivation

We consider just the equations of hydrostatic equilibrium and mass continuity from our equations of stellar structure.

$$\frac{dP}{dr} = -\rho \frac{GM(r)}{r^2}$$

$$\frac{dM}{dr} = 4\pi r^2 \rho$$

We can close the system via an equation of state of the form: $P(\rho)$, which we write as:

$$P = K\rho^{1+1/n}$$

where n is called the *polytropic index*.

With a bit of algebra, we can combine these equations into a single second-order ODE for density:

$$\left(\frac{n+1}{n}\right) \frac{K}{4\pi G} \frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho^{(n-1)/n}} \frac{d\rho}{dr} \right) = -\rho$$

and then make it dimensionless, but expressing the density in terms of the central density, ρ_c :

$$\rho(r) = \rho_c \theta^n(r)$$

where we note that $0 \leq \theta \leq 1$, and a lengthscale α such that $r = \alpha\xi$:

$$\alpha^2 = \frac{(n+1)P_c}{4\pi G \rho_c^2}$$

Giving

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n$$

The boundary conditions are:

- $\theta(\xi = 0) = 1$
- $d\theta/d\xi|_{\xi=0} = 0$ ($g = 0$ at the center, so $dP/dr = d\rho/dr = 0$)

System of first order equations

We will rewrite this as 2 first order equations, taking $y = \theta$, $z = \theta'$:

$$\frac{dy}{d\xi} = z \tag{1}$$

$$\frac{dz}{d\xi} = -\frac{2}{\xi}z - y^n \tag{2}$$

Then we have:

- $y(0) = 1$
- $z(0) = 0$

Notice that if we evaluate the system at $\xi = 0$, then the $dz/d\xi$ term blows up. We need to use an expansion there.

In the limit $\xi \rightarrow 0$, the solution takes the form:

$$\theta(\xi) = 1 - \frac{1}{6}\xi^2 + \frac{n}{120}\xi^4 + \dots$$

which allows us to simplify the $dz/d\xi$ near $\xi = 0$ as:

$$\frac{dz}{d\xi} \approx -\frac{1}{3}$$

Stopping point

But we have the problem that we do not know the stopping point, ξ_1 .

We estimate the radius, ξ_1 each step and make sure that the next step does not take us past that estimate. This prevents us from having negative θ values. Given a point $(\xi_p, \theta(\xi_p))$, and the derivative at that point, $\theta'_p = d\theta/d\xi|_{\xi_p}$, we can write the equation of a line as:

$$\theta(\xi) - \theta_p = \theta'_p(\xi - \xi_p)$$

Then we can ask when does θ become zero, finding:

$$\xi = -\frac{\theta_p}{\theta'_p} + \xi_p$$

or in terms of y and z ,

$$\xi = -\frac{y_p}{z_p} + \xi_p$$

This is our estimate of ξ_1 . We then make sure our stepsize h is small enough that we do not go beyond this estimate.

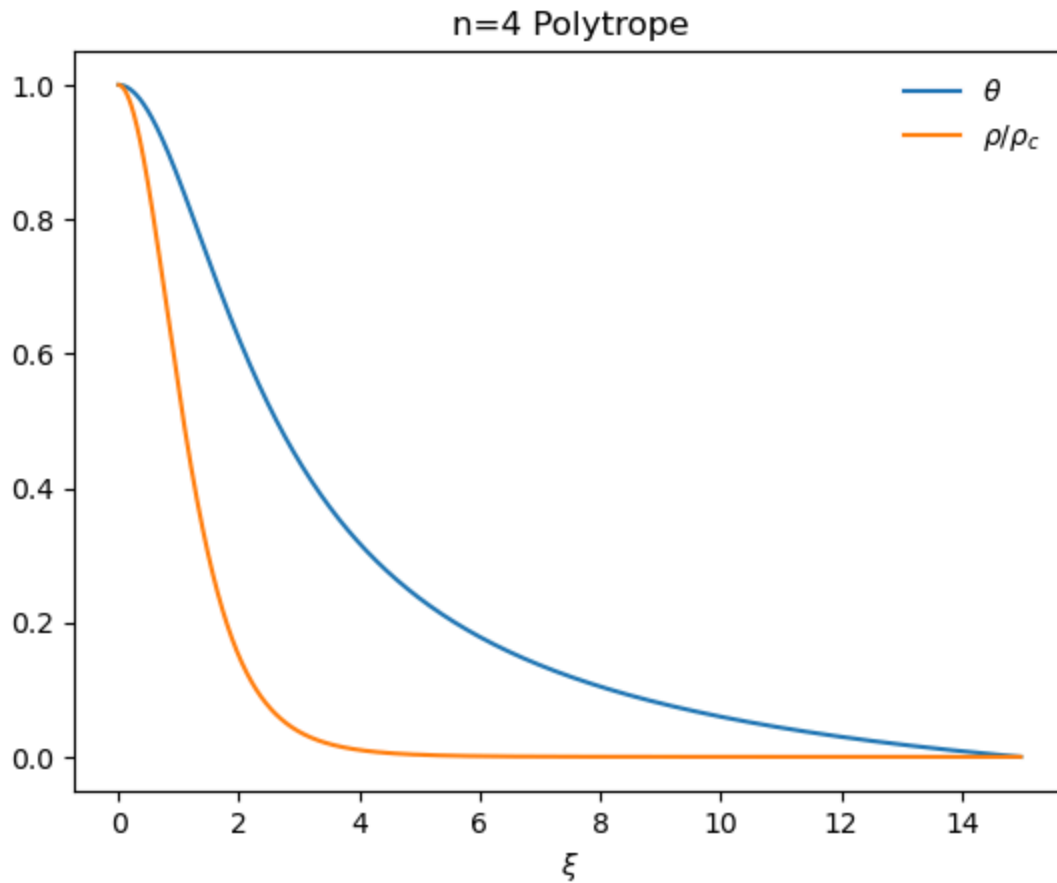
Implementation

This is our main class that does the integration. We initialize it with the polytropic index, and then it will integrate the system for us. We can then plot it or get the parameters ξ_1 and $-\xi_1^2 d\theta/d\xi|_{\xi_1}$

a.

Plot the results of an $n = 4$ polytrope.

```
xi_1 = 14.9715
-xi_1^2 dtheta/dxi|_xi1 = 1.7972
```



From the readings in HKT 7.1-7.2. We found some of the following relations:

Equation 7.20-7.22:

$$\rho(r) = \rho_c \theta^n(r)$$

$$P(r) = P_c \theta^{1+n}(r)$$

$$P_c = K \rho_c^{1+1/n}$$

and

Equation 7.25:

$$r_n^2 = \frac{(n+1)P_c}{4\pi G \rho_c^2}$$

and

$$r = r_n \xi$$

b.

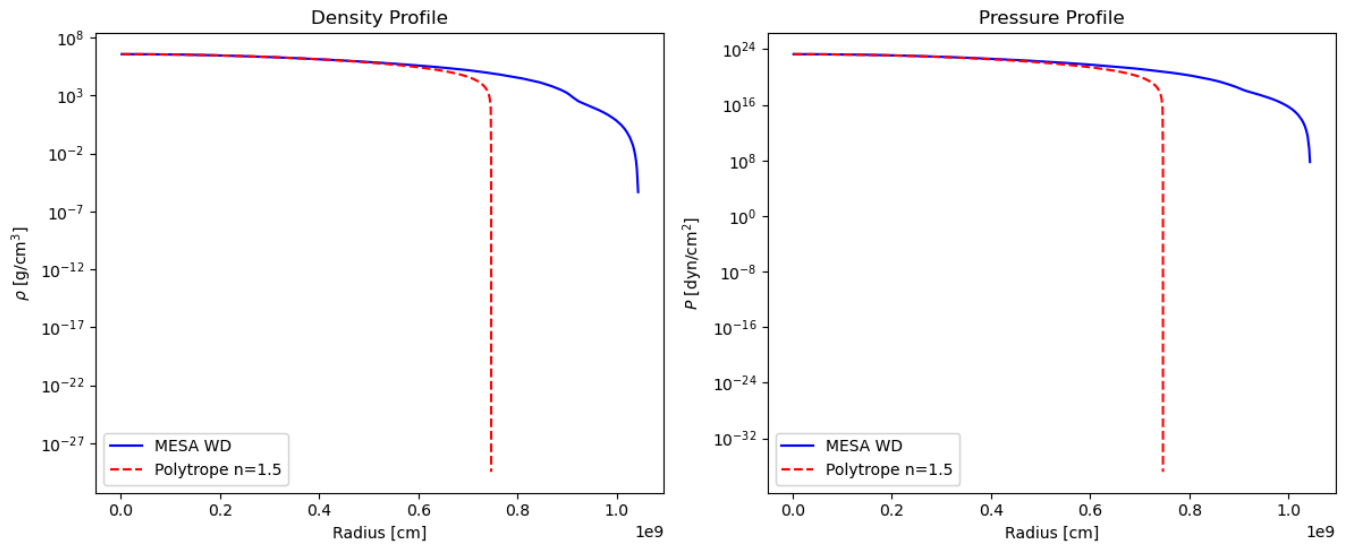
Download the following model files locally. These data were produced using the `make_co_wd` test suite.

- $0.6M_{\odot}$ WD: `co_wd_profile.data`;

Then,

1. produce a polytrope of index `n`
2. compute `K` using the central values from the MESA model (for pandas you can use `iloc`).
3. compute `r_n` using the central values from the MESA model
4. compute the pressure and density profiles from the above constants and the polytrope solutions
5. plot them on the same plot as the MESA model - one plot for pressure one for density
6. try different values of `n` until a best fit is obtained

Central density: $\rho_c = 3.674e+06 \text{ g/cm}^3$
Central pressure: $P_c = 1.891e+23 \text{ dyn/cm}^2$
 $n = 1.5$
 $K = 2.162e+12$
 $r_n = 2.044e+08 \text{ cm}$



C.

Explain in words, the physics motivating the choice of `n` in comparing to the white dwarf.

c result

The best-fit polytropic index is $n = 1.5$, corresponding to a non-relativistic degenerate electron gas where $P \propto \rho^{5/3}$.

This makes physical sense because a $0.6 M_{\odot}$ white dwarf is well below the Chandrasekhar mass limit ($\sim 1.4 M_{\odot}$). At this mass, the central densities are not high enough for the electrons to become relativistic, so the equation of state remains in the non-relativistic degenerate regime.

If the white dwarf were closer to the Chandrasekhar limit, we would expect $n = 3$ (fully relativistic degenerate electrons, $P \propto \rho^{4/3}$) to provide a better fit.

In-Class Assignment: Radially Pulsating Stars - Cepheids

Learning Objectives

- explore boundaries of instability strip
- verify the dominant κ -mechanism source for Cepheids
- compare analytical estimates for the period with MESA output
- compute a phase folded lightcurve to determine changes in pulsation over time.

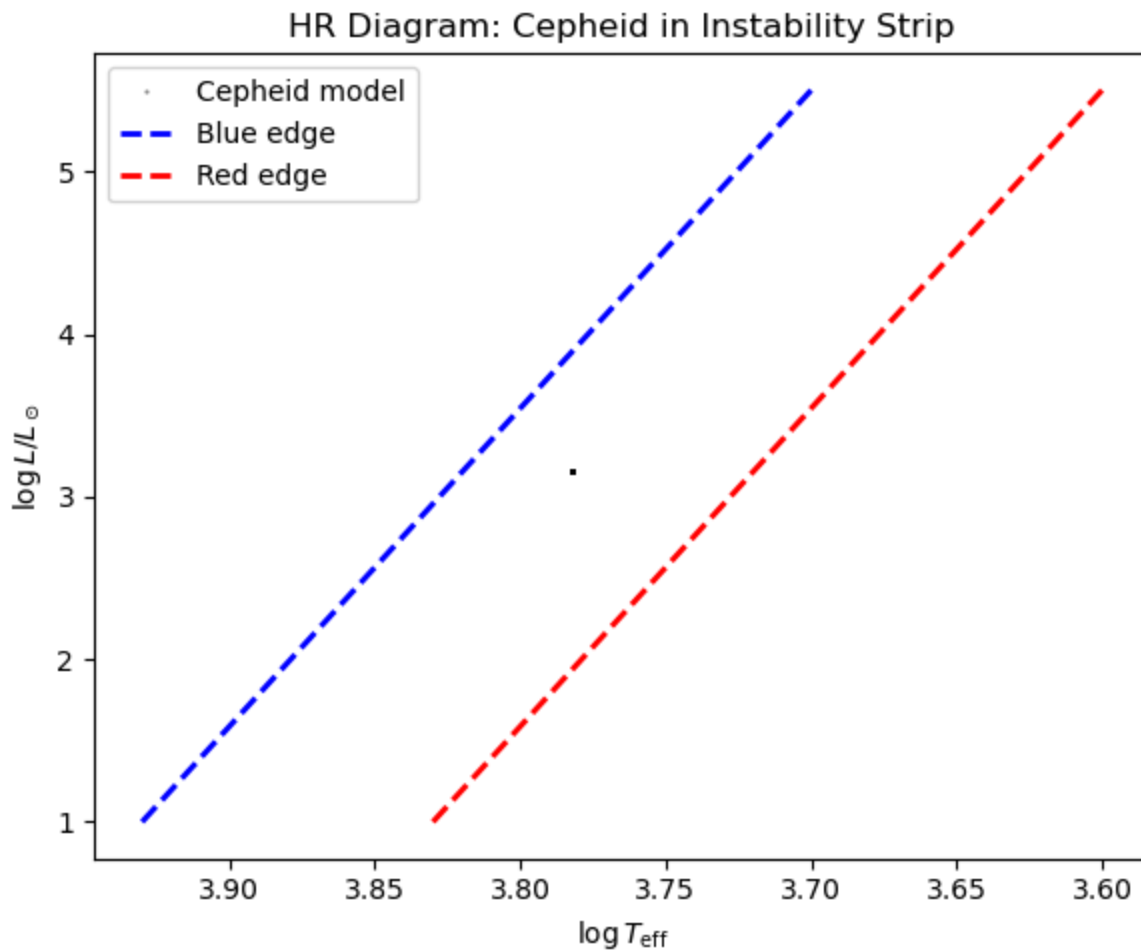
Download the following model files locally.

- Cepheid RSP model: [cephied_history.data](#); [cephied_profile.data](#);

a.

Using the history data,

1. produce an HR diagram of the Cepheid data and confirm that it is bounded within the instability strip.
2. on the same plot, plot the approximate bounds of the instability strip with dashed lines:
 - blue edge point 1 ($\log L_1 = 5.5, \log T_{\text{eff}} = 3.7$);
 - blue edge point 2 ($\log L_2 = 1.0, \log T_{\text{eff}} = 3.93$) and
 - red edge point 1 ($\log L_1 = 5.5, \log T_{\text{eff}} = 3.6$) and
 - red edge point 2 ($\log L_2 = 1.0, \log T_{\text{eff}} = 3.83$).



$\log T_{\text{eff}}$ range: 3.782 to 3.782

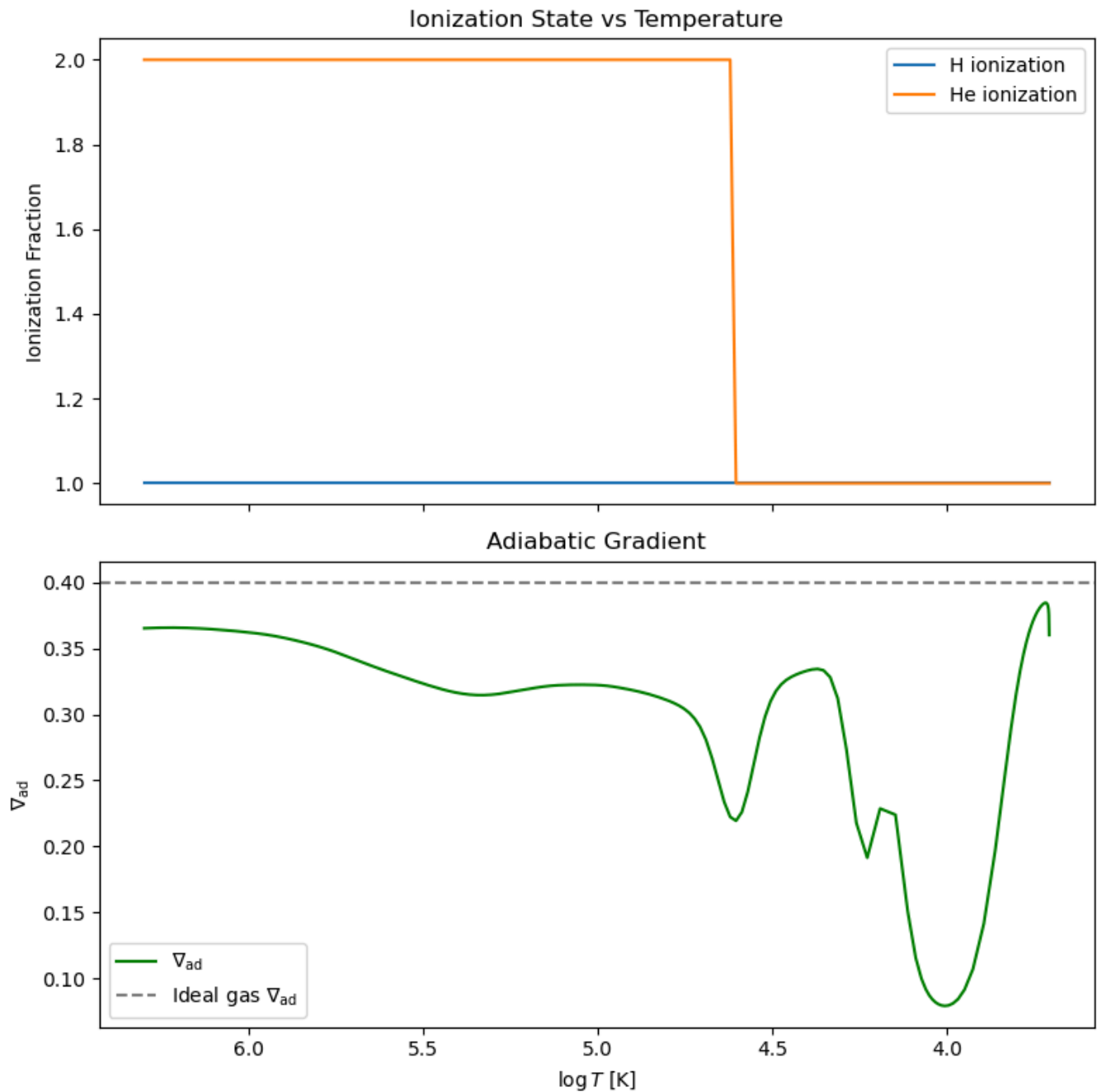
$\log L$ range: 3.157 to 3.159

The model sits within the instability strip boundaries.

b.

Using the profile data,

1. verify the dominant ionization sources via the existence of doubly ionized He and singly ionization H using a plot
2. comment on the location of the ionization zone, that is, is it near the high density region (near the red edge of the strip), or low density (blue edge)
3. plot `grada`, does it decrease as expected in the partial ionization zones?



grada minimum: 0.0793 at $\log T = 4.005$

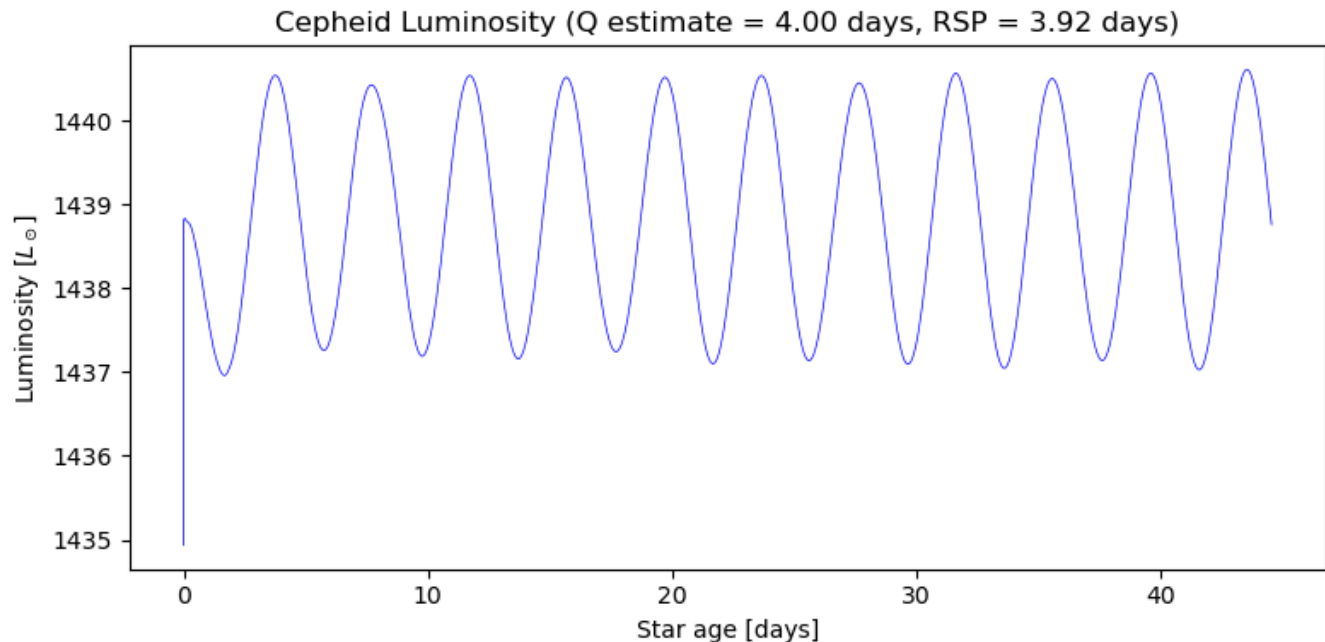
The ionization zones of H and He are visible at $\log T \sim 4.0\text{--}4.2$.
 grada dips significantly in these partial ionization zones,
 falling well below the ideal gas value of 0.4.
 The ionization zone is in a relatively low-density region (outer envelope),
 placing this model closer to the blue edge of the instability strip.

C.

1. Using Eqn. 8.21 in HKT, make an estimate of the period in days. Recall that $\langle \rho \rangle \approx M/R^3$ or Eqn 10.5 in Onno Pol's notes.

2. Plot the total radius or luminosity over time in days `star_age_in_days` , does the estimated period match by eye?

Star mass: 4.165 Msun
Median radius: 34.67 R_{sun}
Mean density: 1.408e-04 g/cm³
Estimated period (full formula): 11.59 days
Estimated period (Q relation): 4.00 days
RSP period (MESA): 3.9240 days



The Q relation gives ~4.0 days, close to the RSP period of 3.92 days.
The full formula overestimates because it assumes uniform density and ideal gas γ_{ad} .

d.

Using the final computed period in the history data `rsp_period_in_days` .

Compute the phase using the period, star age in days, and luminosity.

1. Produce a "phase-folded" luminosity plot - phase vs luminosity.

$$\theta = (t \% P)/P$$

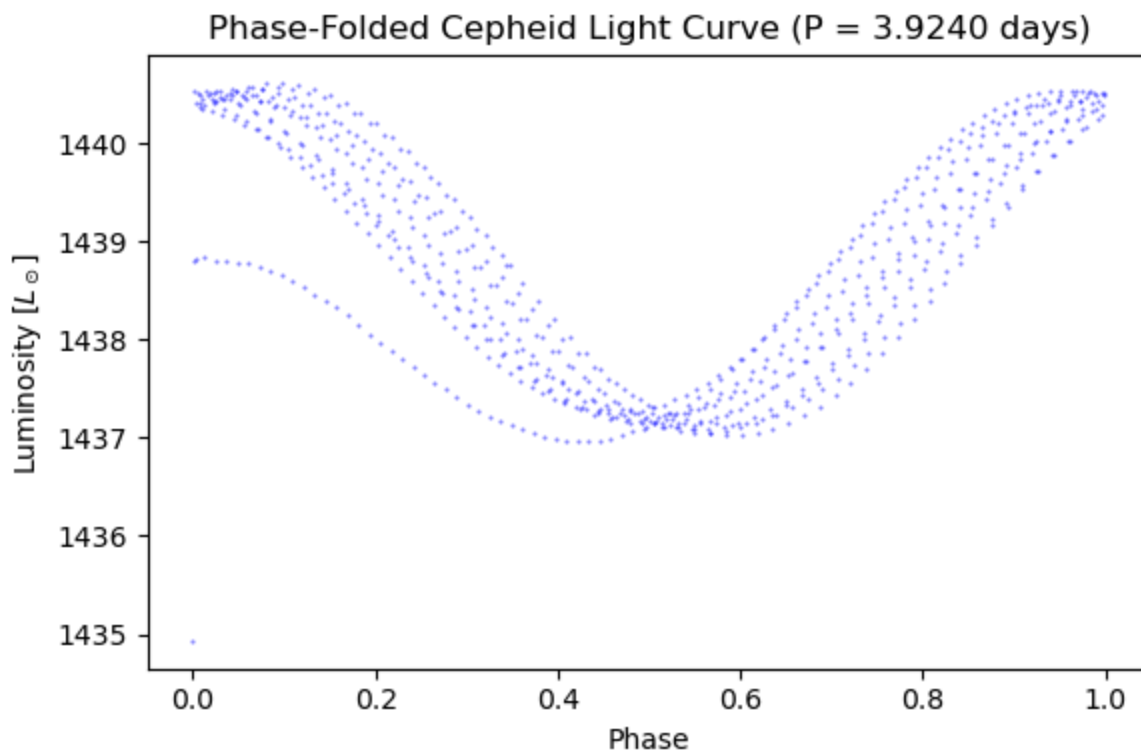
Next, sort by phase to get the argument indices

```
sort_index = np.argsort(phase)

phase_sorted = phase[sort_index]
lum_sorted = lum[sort_index]
```

where `%` is the modulus operator in python. More details and examples are available [here](#) including using Astropy to automatically do the folding.

Comment briefly on if it is possible to observe the growth of the amplitude with subsequent periods. The answer is also in the history data `growth` for each period for the curious.



Growth rate (last period): 0.00881

Growth rates (last 5 values): [0.00823291 0.00823291 0.00823291 0.00823291 0.00881467]

The scatter in the phase-folded plot reflects the growth of the pulsation amplitude over successive periods. The growth column confirms that the amplitude is still increasing (growth > 0).

Rotation

von Zeipel's Paradox

A uniformly rotating star cannot be in steady-state radiative thermal equilibri

Let's start to show this by assuming a cylindrical coordinate system: $\mathbf{r} = \mathbf{r}(\bar{\omega}, \phi, z)$

cylindrical coordinates

- z is aligned with the axis of rotation, "rotation about the Z-axis"
- $\bar{\omega}$ is measured from the z-axis.

We wish to modify the equation of HSE to include the effects of centrifugal force (the apparent outward force on a mass when it is rotated). HSE is our first constraint for such a model.

The first step is to create the total gravitational (or effective potential),

$$\Phi_{\text{eff}}(\bar{\omega}, z) = \Phi_{\text{grav}}(\bar{\omega}, z) - \frac{\Omega^2 \bar{\omega}^2}{2}$$

where $\Phi_{\text{cent}} = -\Omega^2 \bar{\omega}^2 / 2$ and Ω here corresponds to the **angular rotation frequency (rad/s)**.

This leads to our HSE equation now of the form:

$$\frac{1}{\rho} \nabla P = -\nabla \Phi_{\text{eff}} = -\nabla \Phi_{\text{grav}} + \Omega^2 \bar{\omega}^2 \mathbf{e}_{\bar{\omega}} \equiv \mathbf{g}_{\text{eff}}.$$

Our second constraint is for the star to be in thermal balance:

$$\frac{d\mathcal{L}_r}{dr} = 4\pi r^2 \rho \epsilon$$

leading to

$$\nabla \cdot \mathcal{F} = \rho \epsilon$$

where $\mathcal{F}$ now depends on Φ_{eff} .

The final result is that for all of these constraints to be true,

$$\epsilon \propto \left[1 - \frac{\Omega^2}{2\pi G \rho} \right]$$

Some problems with this:

- problem is overconstrained
- energy generation rate *cannot* be directly proportional to rotate rate - unphysical

In-Class Exercise: Effective Gravity on a Rotating Star

Goal

Compute how centrifugal acceleration modifies the effective surface gravity as a function of colatitude θ (measured from the pole).

Background

For a star with mass M , radius R , and angular velocity Ω , the effective gravitational acceleration at colatitude θ is approximately:

$$g_{\text{eff}}(\theta) = \frac{GM}{R^2} - \Omega^2 R \sin^2 \theta$$

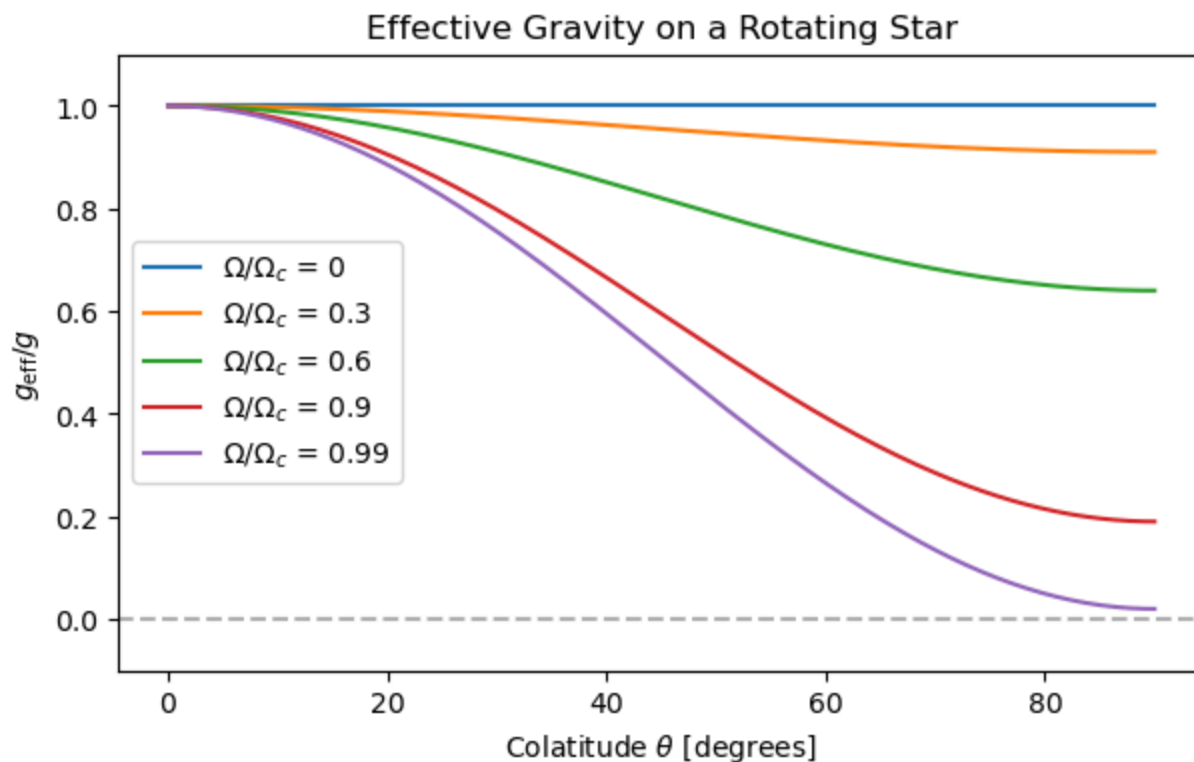
The critical rotation rate, where $g_{\text{eff}} = 0$ at the equator, is:

$$\Omega_{\text{crit}} = \sqrt{\frac{GM}{R^3}}$$

Task

1. Compute g_{eff}/g versus colatitude θ for $\Omega/\Omega_{\text{crit}} = 0, 0.3, 0.6, 0.9, 0.99$.
2. At what colatitude does the effective gravity first reach zero for each rotation rate?
3. What happens to the equatorial gravity as $\Omega \rightarrow \Omega_{\text{crit}}$?

```
<>:11: SyntaxWarning: invalid escape sequence '\0'
<>:11: SyntaxWarning: invalid escape sequence '\0'
/var/folders/fn/00zl2mgj5w1_bhvyzl0w1tp000000gn/T/ipykernel_42289/2582787534.py:11: Sy
ntaxWarning: invalid escape sequence '\0'
plt.plot(np.degrees(theta), g_ratio, label=f'$\Omega/\Omega_c$ = {w}')
```



► Solution

In-Class Exercise: Gravity Darkening Temperature Profile

Goal

Apply von Zeipel's theorem to predict how effective temperature varies across the surface of a rapidly rotating star.

Background

Von Zeipel's theorem states that the local effective temperature scales with the local effective gravity:

$$T_{\text{eff}}(\theta) \propto g_{\text{eff}}(\theta)^\beta$$

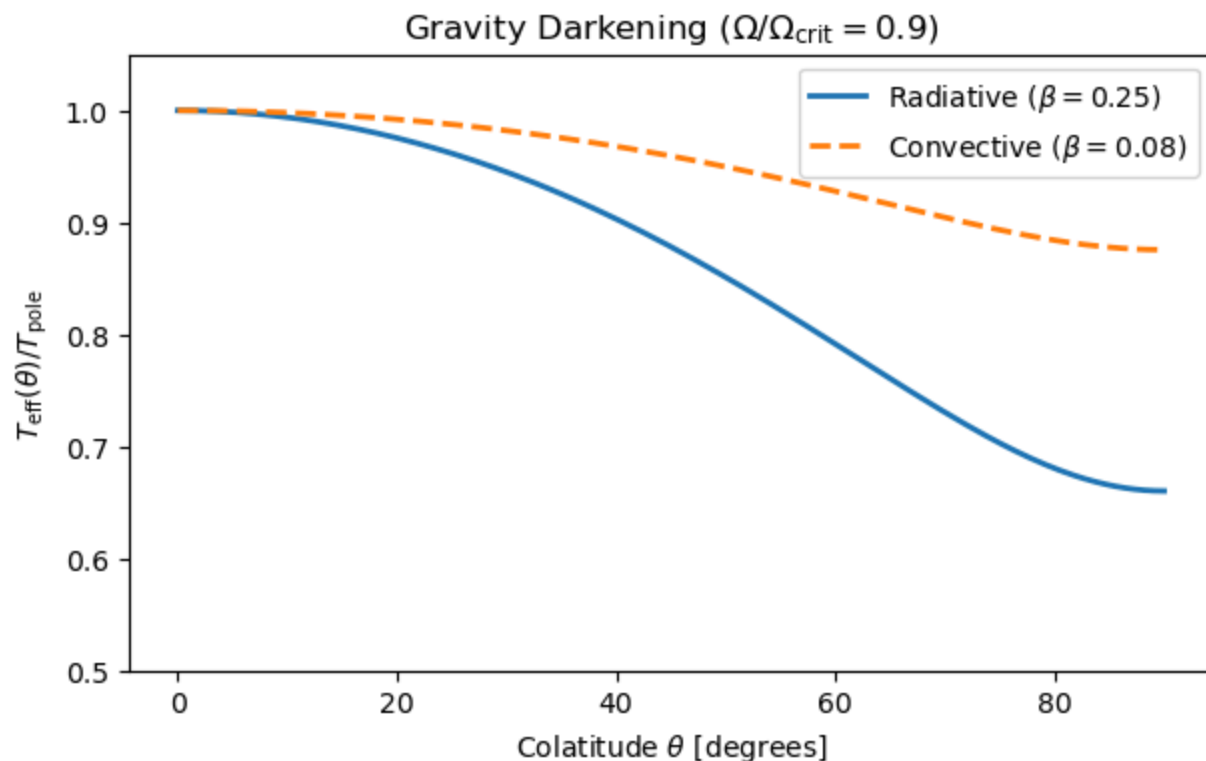
where $\beta = 0.25$ for stars with radiative envelopes (von Zeipel 1924) and $\beta \approx 0.08$ for stars with convective envelopes (Lucy 1967).

Task

1. For $\Omega/\Omega_{\text{crit}} = 0.9$, plot $T_{\text{eff}}(\theta)/T_{\text{pole}}$ versus colatitude for both $\beta = 0.25$ and $\beta = 0.08$.
2. Compute the equator-to-pole temperature ratio for each case.
3. Which type of envelope produces a stronger gravity darkening effect?

beta = 0.25: $T_{\text{eq}}/T_{\text{pole}} = 0.6602$

beta = 0.08: $T_{\text{eq}}/T_{\text{pole}} = 0.8756$



► Solution

The Impact of Rotation of Stellar Evolution

Helioseismology

Helioseismology measurements have led to a few observations about our own Sun:



Caption: Internal rotation in the Sun, showing differential rotation in the outer convective region (as a function of latitude) and almost uniform rotation in the central radiative region. The transition between these regions is called the tachocline. Credit: Global Oscillation Network Group.

1. The Sun is rotating *differentially in the outer convection zone* as a function of latitude.
2. The Sun is rotating close to uniform / solid body in the radiative core.
3. Magnetic fields can help reinforce the strong shear (adjacent layers of fluid moving parallel to each other with different speeds).

Rotation Induced Effects

- Extra mixing, leading enhancement (or destruction) of various elements, bringing material from the core to the surface.
- Extra mass loss, magnetic braking, compact remnant spin rates



Surface abundances at core hydrogen exhaustion as function of the initial stellar mass for models with an equatorial surface rotation rate of 200 km / s at ZAMS. From Heger et al 2000.

In-Class Exercise: Eddington-Sweet Circulation Timescale

Goal

Estimate when meridional circulation (Eddington-Sweet) becomes important compared to the nuclear timescale.

Background

The Eddington-Sweet circulation timescale is approximately:

$$t_{\text{ES}} \approx \frac{t_{\text{KH}}}{(\Omega/\Omega_{\text{crit}})^2}$$

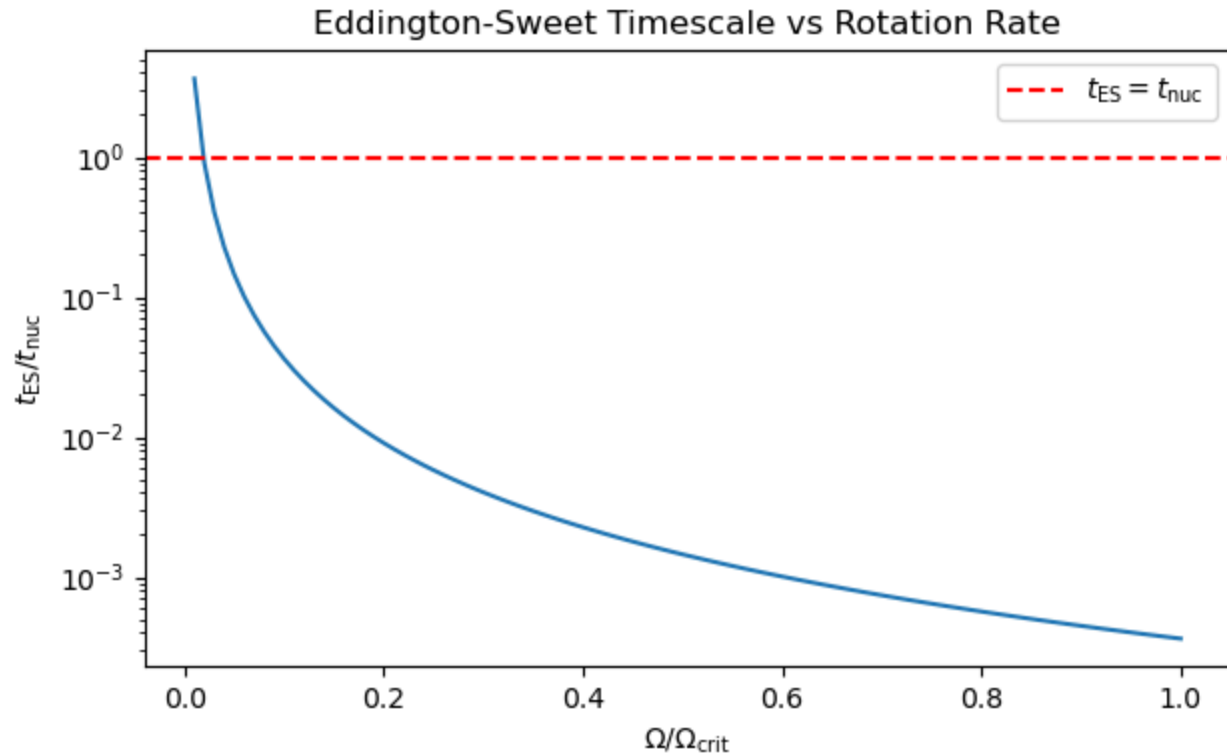
where $t_{\text{KH}} = GM^2/(RL)$ is the Kelvin-Helmholtz (thermal) timescale.

For circulation to significantly alter the star's composition profile during its main-sequence lifetime, we need $t_{\text{ES}} \lesssim t_{\text{nuc}}$.

Task

1. Compute t_{KH} and t_{ES} for a $3 M_{\odot}$ star ($R = 2.5 R_{\odot}$, $L = 100 L_{\odot}$) at several rotation rates.
2. Estimate $t_{\text{nuc}} \approx 0.007 \times Mc^2/L$ (hydrogen burning efficiency $\sim 0.7\%$).
3. Plot $t_{\text{ES}}/t_{\text{nuc}}$ versus $\Omega/\Omega_{\text{crit}}$ and identify the rotation rate above which circulation matters.

$t_{\text{KH}} = 1.132 \times 10^6 \text{ yr}$
 $t_{\text{nuc}} = 3.118 \times 10^9 \text{ yr}$



ES $\sim$ nuc at $\Omega/\Omega_{\text{crit}} \sim 0.02$

► Solution

In-Class Assignment: Gravity Darkening in Low Mass Stars

Gravity darkening, also referred to as gravity brightening, is an astronomical phenomenon where the poles of a star are brighter than the equator, due to rapid rotation and oblate shape. When a star is oblate, it has a larger radius at its equator than it does at its poles.

As a result, the poles have a higher surface gravity, and thus higher temperature and pressure is needed to maintain hydrostatic equilibrium. Thus, the poles are "gravity brightened", and the equator "gravity darkened". - [Wikipedia](#).

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{image}  
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Credit: ESO. Achernar has an unusually rapid rotational velocity, causing it to become oblate in shape.

Learning Objectives

- explore the impact of gravity darkening on HR diagrams
- where does ES operate in rotating stars?
- determine the impact of rotation on the surface carbon abundance

Download the following model files locally.

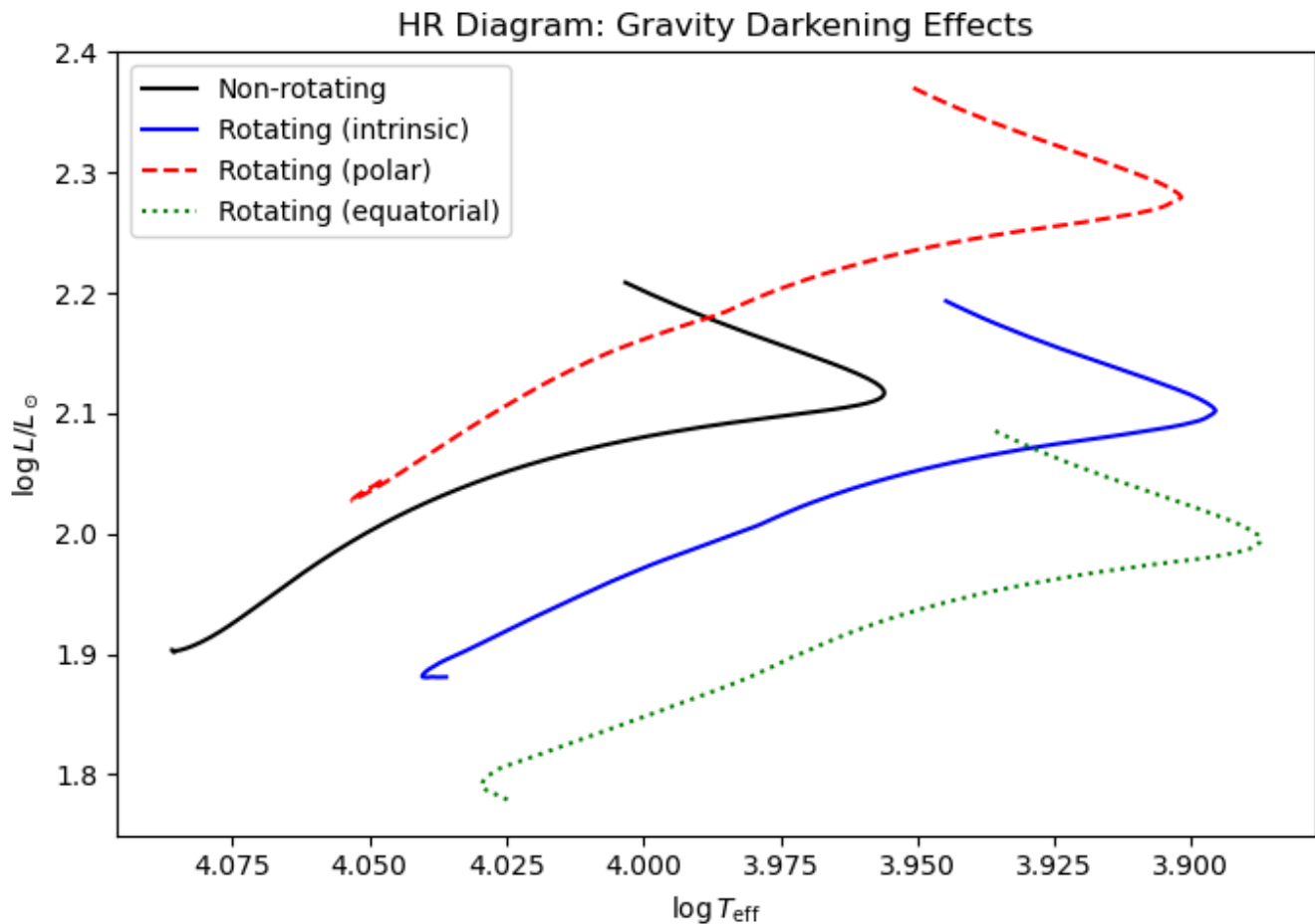
- $3 M_{\odot}$ Model with $(\Omega/\Omega_{\text{crit}})_i = 0.9$: [3m_high_rot_history.data](#);
- $3 M_{\odot}$ Model with $(\Omega/\Omega_{\text{crit}})_i = 0.9$: [3m_high_rot_profile.data](#);
- $3 M_{\odot}$ Model with $(\Omega/\Omega_{\text{crit}})_i = 0.0$: [3m_non_rot_history.data](#);

a. - How does Gravity Darkening Affect HR Evolution?

Using the history data for the rotating and non-rotating models,

1. plot an HR diagram for non-rotating model (our usual history columns) and the rotating model for the (polar, equatorial, and intrinsic - our total usual values).

State how gravity darkening moves the model on the HR, i.e. - hotter/cooler, redder/bluer compared to a non-rotating model.



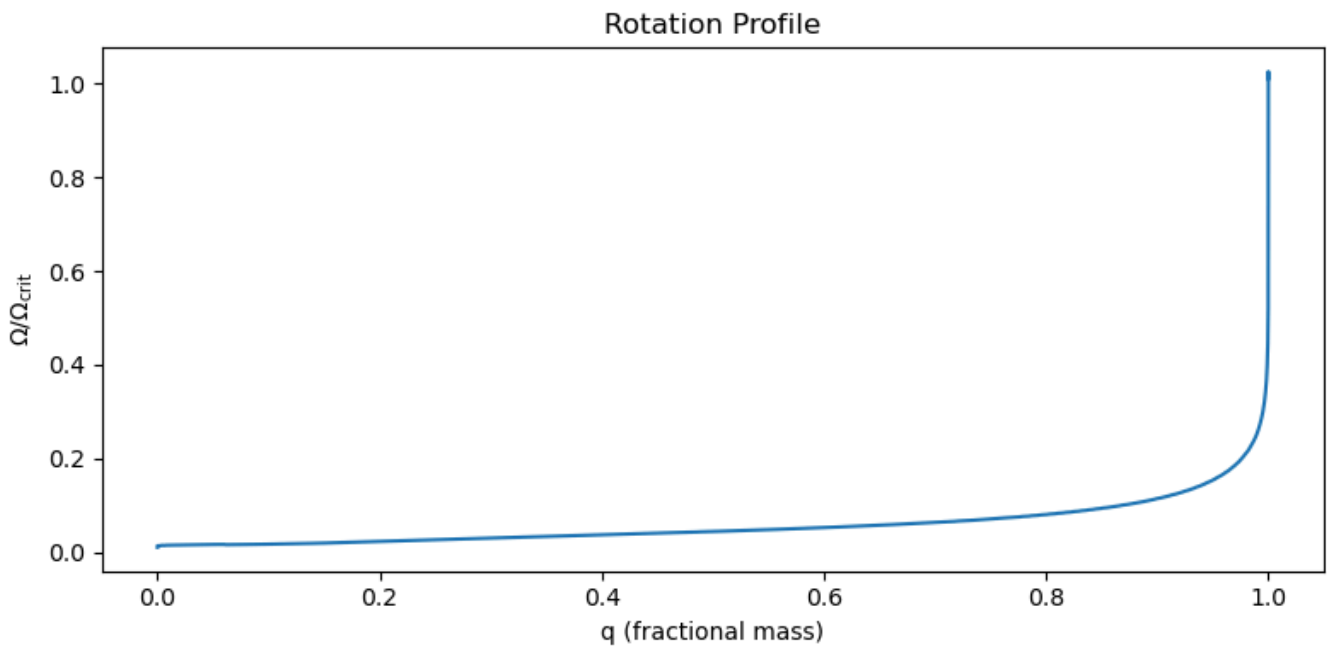
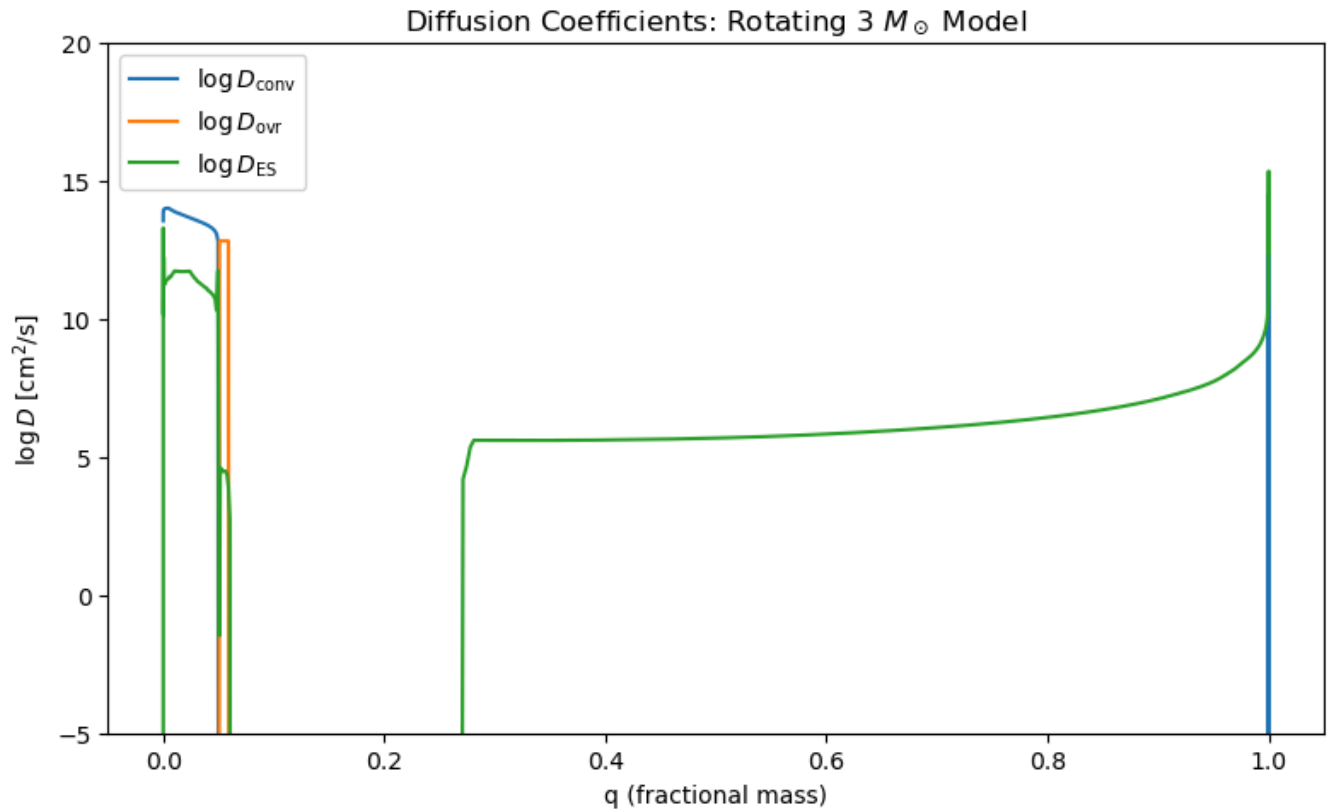
Gravity darkening makes the equator cooler (redder) and dimmer, while the poles appear hotter (bluer) and brighter than the intrinsic values. A pole-on observer sees a hotter, brighter star; an equator-on observer sees cooler, dimmer.

b. - Eddington-Sweet Circulation in Low Mass Rotating Stars

Using the profile data,

1. plot the diffusion coefficient profiles for convection (`log_D_conv`), overshoot (`log_D_ovr`), ES (`am_log_D_ES`) circulation.

Where does the ES circulation operate if anywhere? What can you say about the rotation in these regions?



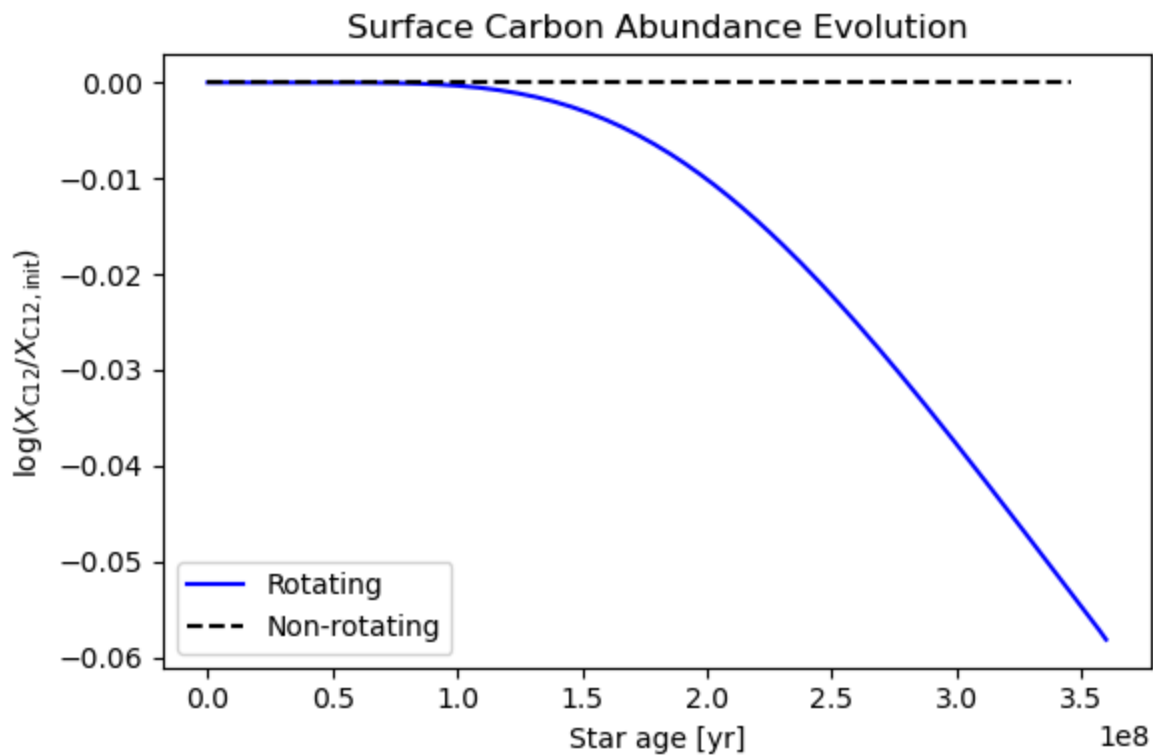
ES circulation operates throughout the radiative zone,
between the convective core and the surface convection zone.
The rotation is differential in these regions, not solid-body.

c. - Impact of Rotation in Surface Carbon Abundance

Using the non-rotating and rotating history data,

1. plot to the log of the initial versus final surface c12 for both models.

For this mass, does rotationally-induced mixing suggest an *increase* or *decrease* to the surface value? Compare with Figure 5 of [Heger et al 2000](#).



Initial C12: 3.4469e-03

Final C12 (rotating): 3.0149e-03 (ratio: 0.8747)

Final C12 (non-rotating): 3.4469e-03 (unchanged)

Rotation causes a DECREASE in surface C12.

Rotationally-induced mixing brings CNO-processed material (depleted in C12, enriched in N14) from the core to the surface. This is consistent with Fig. 5 of Heger et al 2000.